Arithmetic check

Σ BS ~ Σ FS = first RL ~ last RL

6.475 ~ 9.550 = 100.000 ~ 96.925

3.075 = 3.075

Hence checked

(II) Rise and fall method

Station	BS	IS	FS	Rise (+)	Fall (-)	RL	Remarks
A	0.675					100.000	BM
B	0.750		1.230		0.555	99.445	CP_1
C		2.565			1.815	97.630	
D	1.935		2.225	0.340		97.970	CP_2
E		1.835		0.100		98.070	
F	3.115		3.220		1.385	96.685	CP_3
G			2.875	0.240		96.925	
Total	6.475		9.550	0.680	3.755		

Calculation

Level difference of station

Level difference between the stations A and B = 0.675 – 1.230 = - 0.555

Level difference between the stations B and C = 0.750 – 2.565 = - 1.815

Level difference between the stations C and D = 2.565 – 2.225 = - 1.960

Level difference between the stations D and E = 1.935 – 1.835 = + 0.100

Level difference between the stations E and F = 1.835 – 3.220 = - 1.285

Level difference between the stations F and G = 3.115 – 2.875 = + 0.240

RL of station

RL of B = RL of A ± level difference of B = 100.00 – 0.555 = 99.445

RL of C = RL of B ± level difference of C = 99.445 – 1.815 = 97.630

RL of D = RL of C ± level difference of D = 97.630 +0.340 = 97.970

RL of E = RL of D ± level difference of E = 97.970 + 0.100 = 98.070

RL of F = RL of E ± level difference of F = 98.070 – 1.385 = 96.685

RL of G = RL of F ± level difference of G = 96.685 + 0.240 = 96.925

Arithmetic check

Σ BS ~ Σ FS = Σ Rise ~ Σ fall = first RL ~ last RL

6.475 ~ 9.550 = 0.680 ~ 3.755 = 100.000 ~ 96.925

3.075 = 3.075 = 3.075

Hence checked

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